

Problem 1

Solution:

Theorem 1.5 $a(b - c) = ab - ac$.

$$\begin{aligned}
 a(b - c) + ac &= a((b - c) + c) && \text{(by distributive law)} \\
 &= a((b + (-c)) + c) && \text{(by theorem 1.3)} \\
 &= a(b + ((-c) + c)) && \text{(by associative law)} \\
 &= a(b + (c + (-c))) && \text{(by commutative law)}
 \end{aligned}$$

By axiom 5, there exists an x such that $c + x = 0$. And by theorem 1.2, we know that this x is $0 - c = -c$. Hence, $c + (-c) = 0$ so

$$\begin{aligned}
 a(b - c) + ac &= a(b + 0) \\
 &= ab && \text{(by axiom 4)} \\
 ac + a(b - c) &= ab && \text{(by commutative law)} \\
 a(b - c) &= ab - ac && \text{(by theorem 1.2)}
 \end{aligned}$$

which is what we wanted to show.

Theorem 1.6 $0 \cdot a = a \cdot 0 = 0$.

By the commutative law, $0 \cdot a = a \cdot 0$. So all that's left is to show $a \cdot 0 = 0$.

$$\begin{aligned}
 a \cdot 0 &= a(c + (-c)) && \text{(by axiom 5 and theorem 1.2)} \\
 &= a(c - c) && \text{(by theorem 1.3)} \\
 &= ac - ac && \text{(by theorem 1.5)} \\
 &= ac + (-ac) && \text{(by theorem 1.3)} \\
 &= 0 && \text{(by axiom 5 and theorem 1.2)}
 \end{aligned}$$

as desired.

Theorem 1.7 Cancellation law for multiplication. If $ab = ac$ and $a \neq 0$ then $b = c$. In particular, this shows that the number 1 of axiom 4 is unique.

By axiom 6, since $a \neq 0 \exists y$ such that $ay = 1$. And $ab = ac \implies y(ab) = y(ac)$, so

$$\begin{aligned}
 y(ab) &= y(ac) \\
 (ya)b &= y(ac) && \text{(by associative law)} \\
 (ay)b &= y(ac) && \text{(by commutative law)} \\
 1b &= y(ac) \\
 b &= y(ac) && \text{(by axiom 4)} \\
 b &= (ya)c && \text{(by associative law)} \\
 b &= (ay)c && \text{(by commutative law)} \\
 b &= c && \text{(by axiom 4)}
 \end{aligned}$$

For the second part, we must show that 1 from axiom 4 is unique. For this, suppose that there are two 1s, 1 and 1' and that $1 \cdot x = 1' \cdot x$ where $x \neq 0$. We can rewrite this as $x \cdot 1 = x \cdot 1'$ by commutative law. Now, observe that these are precisely the conditions under which the first half of this theorem applies, so we can immediately conclude that $1 = 1'$.

Theorem 1.8 Possibility of division. Given a, b with $a \neq 0$ there is exactly one x such that $ax = b$. This x is denoted by $\frac{b}{a}$ and is called the quotient of b and a . In particular, $\frac{1}{a}$ is also written a^{-1} and is called the reciprocal of a .

By axiom 6, since $a \neq 0$ there exists y such that $ay = 1$. Now,

$$\begin{aligned}
 (ay)b &= 1b \\
 &= b && \text{(by axiom 4)} \\
 a(yb) &= b && \text{(by associativity law)}
 \end{aligned}$$

so we have shown that $x = yb$ is one possible x . To show that this x is unique, suppose there two possible numbers — x and x' — and $ax = ax' = b$. But by theorem 1.7, since $a \neq 0$, we can immediately conclude $x = x'$.

Theorem 1.9 If $a \neq 0$, then $\frac{b}{a} = ba^{-1}$

Let $x = \frac{b}{a} \implies ax = b$. Similarly, let $y = a^{-1}$. Then, $y = \frac{1}{a} \implies ay = 1$

$$\begin{aligned}
 ax &= (1)b && \text{(by axiom 4)} \\
 &= (ay)b \\
 &= a(yb) && \text{(by associativity law)} \\
 &= a(by) && \text{(by commutativity law)} \\
 \implies x &= by && \text{(by theorem 1.7)} \\
 \implies \frac{b}{a} &= ba^{-1}
 \end{aligned}$$

Theorem 1.10 If $a \neq 0$ then $(a^{-1})^{-1} = a$.

By theorem 1.8 $a \cdot a^{-1} = 1$. And $a^{-1} \neq 0$ because otherwise, by theorem 1.6 $a \cdot a^{-1} = a \cdot 0 = 0$ (and we know by axiom 4 that $0 \neq 1$). Now, by commutative law, $a^{-1} \cdot a = 1$ as well. But by theorem 1.8, we can rewrite this as $a^{-1} \cdot (a^{-1})^{-1} = 1$ (and it is defined because we just showed that $a^{-1} \neq 0$). So now we have $a^{-1} \cdot a = a^{-1} \cdot (a^{-1})^{-1}$ and $a^{-1} \neq 0$, which allows us to simplify to $a = (a^{-1})^{-1}$ by theorem 1.7.

Theorem 1.11 If $ab = 0$ then either $a = 0$ or $b = 0$.

We will show this by contradiction. Suppose $ab = 0$ but $a \neq 0$ and $b \neq 0$. Since $a \neq 0$, there exists an a^{-1} such that $a \cdot a^{-1} = 1$ by axiom 6 and theorem 1.8. Now,

$$\begin{aligned}
 b &= 1b && \text{(by axiom 4)} \\
 &= (a \cdot a^{-1})b \\
 &= (a^{-1} \cdot a)b && \text{(by commutative law)} \\
 &= a^{-1}(ab) && \text{(by associative law)} \\
 &= a^{-1}(0) \\
 &= 0 && \text{(by theorem 1.6)}
 \end{aligned}$$

But this is a contradiction since we supposed $b \neq 0$.

Theorem 1.12 $(-a)b = -ab$ and $(-a)(-b) = ab$.

$$\begin{aligned}
 (-a)b &= (0 - a)b && \text{(by theorem 1.2)} \\
 &= b(0 - a) && \text{(by commutative law)} \\
 &= b(0) - ba && \text{(by theorem 1.5)} \\
 &= 0 - ba && \text{(by theorem 1.6)} \\
 &= 0 - ab && \text{(by commutative law)} \\
 &= -ab && \text{(by theorem 1.2)}
 \end{aligned}$$

$$\begin{aligned}
 (-a)(-b) &= (-a)(0 - b) && \text{(by theorem 1.2)} \\
 &= (-a)0 - (-a)b && \text{(by commutative law)} \\
 &= 0 - (-a)b && \text{(by theorem 1.6)} \\
 &= 0 - (-ab) && \text{(by first part of theorem 1.12)} \\
 &= -(-ab) && \text{(by theorem 1.2)} \\
 &= ab && \text{(by theorem 1.4)}
 \end{aligned}$$

Theorem 1.13 If $b \neq 0$ and $d \neq 0$, then $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$

Lemma: bd has reciprocal $(bd)^{-1}$.

Proof: by contrapositive of theorem 1.11, since $b \neq 0$ and $d \neq 0$, $bd \neq 0$. Therefore, we can invoke theorem 1.8 and conclude that bd has a reciprocal $(bd)^{-1}$, i.e. $(bd)(bd)^{-1} = 1$

$$\begin{aligned}
 \frac{a}{b} + \frac{c}{d} &= 1 \left(\frac{a}{b} + \frac{c}{d} \right) && \text{(by axiom 4)} \\
 &= ((bd)(bd)^{-1}) \left(\frac{a}{b} + \frac{c}{d} \right) && \text{(by lemma)} \\
 &= ((bd)^{-1}(bd)) \left(\frac{a}{b} + \frac{c}{d} \right) && \text{(by commutativity law)} \\
 &= (bd)^{-1} \left((bd) \left(\frac{a}{b} + \frac{c}{d} \right) \right) && \text{(by associative law)} \\
 &= (bd)^{-1} ((bd)(ab^{-1} + cd^{-1})) && \text{(by theorem 1.9)} \\
 &= (bd)^{-1} ((bd)(ab^{-1}) + (bd)(cd^{-1})) && \text{(by distributive law)} \\
 &= (bd)^{-1} ((ab^{-1})(bd) + (bd)(cd^{-1})) && \text{(by commutative law)} \\
 &= (bd)^{-1} ((ab^{-1})(bd) + (bd)(d^{-1}c)) && \text{(by commutative law)} \\
 &= (bd)^{-1} (a(b^{-1}b)d + (bd)(d^{-1}c)) && \text{(by associative law)} \\
 &= (bd)^{-1} (a(b^{-1}b)d + b(dd^{-1})c) && \text{(by associative law)} \\
 &= (bd)^{-1} (a(bb^{-1})d + b(dd^{-1})c) && \text{(by commutative law)} \\
 &= (bd)^{-1} (a(1)d + b(1)c) && \text{(by theorem 1.8)} \\
 &= (bd)^{-1} (a((1)d) + b(1)c) && \text{(by associative law)} \\
 &= (bd)^{-1} (a((1)d) + b((1)c)) && \text{(by associative law)} \\
 &= (bd)^{-1} (ad + b((1)c)) && \text{(by axiom 4)} \\
 &= (bd)^{-1} (ad + bc) && \text{(by axiom 4)} \\
 &= (ad + bc)(bd)^{-1} && \text{(by commutative law)} \\
 &= \frac{ad + bc}{bd} && \text{(by theorem 1.9)}
 \end{aligned}$$

Theorem 1.14 If $b \neq 0$ and $d \neq 0$, then $\left(\frac{a}{b}\right)\left(\frac{c}{d}\right) = \frac{ac}{bd}$

Lemma: If $b \neq 0$ and $d \neq 0$ then $b^{-1}d^{-1} = (bd)^{-1}$.

Proof: By the lemma from theorem 1.13, we know that bd has a reciprocal $(bd)^{-1}$, i.e. $bd(bd)^{-1} = 1$. Then, $(b^{-1}d^{-1})(bd)(bd)^{-1} = (b^{-1}d^{-1})1$. We'll simplify the left-hand side first

$$\begin{aligned}
 (b^{-1}d^{-1})(bd)(bd)^{-1} &= (b^{-1}(d^{-1}b)d)(bd)^{-1} && \text{(by associative law)} \\
 &= (b^{-1}(bd^{-1})d)(bd)^{-1} && \text{(by commutative law)} \\
 &= ((b^{-1}b)(d^{-1}d))(bd)^{-1} && \text{(by associative law)} \\
 &= ((1)(d^{-1}d))(bd)^{-1} && \text{(by theorem 1.8)} \\
 &= ((1)(1))(bd)^{-1} && \text{(by theorem 1.8)} \\
 &= (1)(bd)^{-1} && \text{(by axiom 4)} \\
 &= (bd)^{-1} && \text{(by axiom 4)}
 \end{aligned}$$

followed by the righthand side.

$$\begin{aligned}
 (b^{-1}d^{-1})1 &= 1((b^{-1}d^{-1})) && \text{(by commutative law)} \\
 &= b^{-1}d^{-1} && \text{(by axiom 4)}
 \end{aligned}$$

Thus, we have $(bd)^{-1} = b^{-1}d^{-1}$ as desired. Now, observe that

$$\begin{aligned}
 \left(\frac{a}{b}\right)\left(\frac{c}{d}\right) &= (ab^{-1}(cd^{-1})) && \text{(by theorem 1.9)} \\
 &= (ab^{-1}(d^{-1}c)) && \text{(by commutative law)} \\
 &= (ab^{-1}d^{-1})c && \text{(by associative law)} \\
 &= (a(b^{-1}d^{-1}))c && \text{(by associative law)} \\
 &= ((b^{-1}d^{-1})a)c && \text{(by commutative law)} \\
 &= (b^{-1}d^{-1})(ac) && \text{(by associative law)} \\
 &= (bd)^{-1}(ac) && \text{(by lemma)} \\
 &= (ac)(bd)^{-1} && \text{(by commutative law)} \\
 &= \frac{ac}{bd} && \text{(by theorem 1.9)}
 \end{aligned}$$

Theorem 1.15 If $b \neq 0$ and $c \neq 0$ and $d \neq 0$, then $\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{ad}{bc}$

Lemma: cd^{-1} has a reciprocal $(cd^{-1})^{-1}$, i.e. $cd^{-1}((cd^{-1})^{-1}) = 1$.

Proof: First, observe that $d^{-1} \neq 0$. Otherwise, $dd^{-1} = d(0) = 0$ (by theorem 1.6) but we by theorem 1.8 that since $d \neq 0$, d^{-1} is well-defined, i.e. $dd^{-1} = 1$. Now, since $c \neq 0$ and $d^{-1} \neq 0$, we know by the contrapositive of theorem 1.11 that $cd^{-1} \neq 0$. Therefore, we can invoke theorem 1.8 to conclude that $(cd^{-1})^{-1}$ is the reciprocal of cd^{-1} .

$$\begin{aligned}
 \frac{\frac{a}{b}}{\frac{c}{d}} &= \frac{ab^{-1}}{\frac{c}{d}} && \text{(by theorem 1.9)} \\
 &= \frac{ab^{-1}}{cd^{-1}} && \text{(by theorem 1.9)} \\
 &= (ab^{-1})(cd^{-1})^{-1} && \text{(by lemma)} \\
 &= (ab^{-1})(c^{-1}(d^{-1})^{-1}) && \text{(by lemma of theorem 1.14)} \\
 &= (ab^{-1})(c^{-1}d) && \text{(by theorem 1.10)} \\
 &= a(b^{-1}c^{-1})d && \text{(by associative law)} \\
 &= a((bc)^{-1})d && \text{(by lemma of theorem 1.14)} \\
 &= a((bc)^{-1}d) && \text{(by associative law)} \\
 &= a(d(bc)^{-1}) && \text{(by commutative law)} \\
 &= (ad)(bc)^{-1} && \text{(by associative law)} \\
 &= \frac{ad}{bc} && \text{(by theorem 1.9)}
 \end{aligned}$$

□

Problem 2

Solution:

Let $x = -0$. By theorem 1.2, $x = 0 - 0$, i.e. it is the real number such that $0 + x = 0$. However, by the commutative law $0 + x = x + 0$ and by axiom 4, $x + 0 = x$. Therefore, $0 + x = x = 0$, so we can conclude $-0 = 0$.

□

Problem 3

Solution:

By axiom 4, $1 \neq 0$. Therefore, by theorem 1.8, 1 has a reciprocal 1^{-1} such that $1 \cdot 1^{-1} = 1$. But by axiom 4, $1 \cdot 1^{-1}$ simplifies to 1^{-1} so we can conclude that $1^{-1} = 1$.

□

Problem 4

Solution:

Suppose by way of contradiction that 0 has a reciprocal, i.e. there exists a 0^{-1} such that $0 \cdot 0^{-1} = 1$. However, by theorem 1.6, we know that $0 \cdot 0^{-1} = 0$ and by axiom 4, we know that $0 \neq 1$. This a contradiction. Thus, 0 has no reciprocal.

□

Problem 5

Solution:

By theorem 1.2 $a + (-a) = 0$ and $b + (-b) = 0$. And $(a + (-a) + (b + (-b))) = 0 + 0 = 0$ by axiom 4. Now, the lefthand side can be rewritten as

$$\begin{aligned}
 (a + (-a)) + (b + (-b)) &= (a + (-a)) + ((-b) + b) && \text{(by commutative law)} \\
 &= a + ((-a) + (-b)) + b && \text{(by associative law)} \\
 &= a + (((-a) + (-b)) + b) && \text{(by associative law)} \\
 &= a + (b + ((-a) + (-b))) && \text{(by commutative law)} \\
 &= (a + b) + ((-a) + (-b)) && \text{(by associative law)} \\
 &= (a + b) + (-a - b) && \text{(by theorem 1.3)}
 \end{aligned}$$

so we have $(a+b)+(-a-b) = 0$. However, by theorem 1.8 it's also true that $(a+b)+-(a+b) = 0$. Hence, we have $(a + b) + (-a - b) = (a + b) + -(a + b)$. By theorem 1.1, this implies that $-a - b = -(a + b)$.

□

Problem 6

Solution:

$$\begin{aligned}
 -(a - b) &= -(a + (-b)) && \text{(by theorem 1.3)} \\
 &= -a - (-b) && \text{(by problem 5)} \\
 &= -a + b && \text{(by theorem 1.4)}
 \end{aligned}$$

□

Problem 7

Solution:

$$\begin{aligned}
 (a - b) + (b - c) &= (a + (-b)) + (b - c) && \text{(by theorem 1.3)} \\
 &= (a + (-b)) + (b + (-c)) && \text{(by theorem 1.3)} \\
 &= a + ((-b) + b) + (-c) && \text{(by associative law)} \\
 &= a + (b + (-b)) + (-c) && \text{(by commutative law)} \\
 &= a + 0 + (-c) && \text{(by theorem 1.2)} \\
 &= (a + 0) + (-c) && \text{(by associative law)} \\
 &= a + (-c) && \text{(by axiom 4)} \\
 &= a - c && \text{(by theorem 1.3)}
 \end{aligned}$$

□

Problem 8

Solution:

We already proved this as a lemma in theorem 1.14

□

Problem 9

Solution:

$$\begin{aligned}
 -\left(\frac{a}{b}\right) &= -(ab^{-1}) && \text{(by theorem 1.9)} \\
 &= (-a)b^{-1} && \text{(by theorem 1.12)} \\
 &= \frac{(-a)}{b} && \text{(by theorem 1.9)}
 \end{aligned}$$

Lemma: If $b \neq 0$, then $(-b)^{-1} = -(b^{-1})$

Proof: We first need to show that $b \neq 0 \implies (-b) \neq 0$. We'll do it by contradiction. Suppose that $-b = 0$. By theorem 1.2 $b + (-b) = 0$ so $b + 0 = 0$. But by axiom 4, we know that $b + 0 = b$, so we have $b = 0$, which is a contradiction. Hence, $(-b) \neq 0$.

Since $-b \neq 0$, it has, by theorem 1.8, a reciprocal $(-b)^{-1}$ such that $(-b)((-b)^{-1}) = 1$. Likewise, b has a reciprocal b^{-1} such that $b(b^{-1}) = 1$.

Therefore, given theorem 1.12, we can say that $(-b)(-(b^{-1})) = bb^{-1} = 1$.

Putting both pieces together, we arrive at $(-b)((-b)^{-1}) = (-b)(-(b^{-1})) = 1$. And since we've already shown that $-b \neq 0$, we can invoke theorem 1.7 to conclude that $(-b)^{-1} = -(b^{-1})$.

$$\begin{aligned}
 -\left(\frac{a}{b}\right) &= -(ab^{-1}) && \text{(by theorem 1.9)} \\
 &= -(b^{-1}a) && \text{(by commutative law)} \\
 &= (-b^{-1})a && \text{(by theorem 1.12)} \\
 &= ((-b)^{-1})a && \text{(by lemma)} \\
 &= a((-b)^{-1}) && \text{(by commutative law)} \\
 &= \frac{a}{(-b)} && \text{(by theorem 1.9)}
 \end{aligned}$$

□

Problem 10

Solution:

$$\begin{aligned}\left(\frac{a}{b}\right) - \left(\frac{c}{d}\right) &= \left(\frac{a}{b}\right) + \left(-\left(\frac{c}{d}\right)\right) && \text{(by theorem 1.3)} \\ &= \left(\frac{a}{b}\right) + \left(\frac{(-c)}{d}\right) && \text{(by problem 9)} \\ &= \frac{ad + b(-c)}{bd} && \text{(by theorem 1.13)} \\ &= \frac{ad + (-c)b}{bd} && \text{(by commutative law)} \\ &= \frac{ad + -(cb)}{bd} && \text{(by theorem 1.12)} \\ &= \frac{ad + -(bc)}{bd} && \text{(by commutative law)} \\ &= \frac{ad - bc}{bd} && \text{(by theorem 1.3)}\end{aligned}$$

□